Objective:

Demonstrate a comprehensive understanding of deep generative models by implementing Variational Autoencoder (VAE) and Denoising Diffusion Probabilistic Model (DDPM) architectures from the ground up., showcasing creativity and technical understanding of generative models.

Overview:

Variational Autoencoders (VAEs) and Denoising Diffusion Probabilistic Models (DDPMs) are both prominent architectures for image generation. Each has its own unique behaviour, architecture, and underlying logic. Although DDPMs are often considered the state-of-the-art for high-quality image synthesis, both models have their own strengths and weaknesses.

- **Variational Autoencoders (VAEs)**

VAEs are trained by mapping input images into a latent space, which is encouraged to resemble a Gaussian distribution, and then reconstructing the images from that latent space. To achieve these objectives, the VAE loss function consists of two terms:

1. **Reconstruction loss**, which ensures that the output preserves the features and structure of the input image.
2. **KL divergence loss**, which ensures that the latent space follows a Gaussian distribution. This alignment allows random latent vectors sampled at inference to produce meaningful outputs rather than "garbage" images.

The Beta-VAE is a variation that introduces a hyperparameter **β** to the loss function, creating a trade-off between reconstruction quality and latent space regularization. A smaller β emphasizes reconstruction quality, which can produce sharper images but risks generating unrealistic outputs if the latent vector does not align with the learned Gaussian structure. A larger β emphasizes latent space regularization, reducing hallucinations but potentially compromising reconstruction quality.

- **VAE Loss:**

$$L_{VAE} = Reconstruction\ Loss\ (MSE) + \beta * KL\ Divergence$$

$$L_{VAE} = E_{q\phi(Z|X)}[\| x - \hat{x} \|^2] + \beta * D_{KL}(q\phi(z \mid x) \parallel p(z))$$

- x is the input image

- $\hat{x}$ is the reconstructed image
- z is the latent vector sampled from $q_\phi(z \mid x)$
- $p(z)$ is the prior Gaussian distribution
- β balances reconstruction vs latent space regularization

Despite its simplicity and intuitive design, the VAE suffers from a major limitation: its generated images are often blurry. This occurs because the MSE loss used for reconstruction encourages the model to predict the mean of the output distribution at each pixel. While this minimizes overall error and produces realistic-looking images, it smooths out edges and fine details, resulting in blurred boundaries.

- **Denoising Diffusion Probabilistic Models (DDPMs)**

DDPMs are more sophisticated architectures than VAEs. First, an image is taken as input, and a specific amount of noise is added to that image using a predefined schedule. The noise amount is a function of the timestep, so a larger timestep results in a more deteriorated image. The process is formulated such that as the timestep increases, the deteriorated image approaches pure Gaussian noise with a mean of zero and a standard deviation of one; this is known as the forward pass.

- **Forward process (adding noise step by step):**

At each timestep t, noise is added to the image $x_0$to produce x_t:

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\,\text{noise, where noise } \sim \text{Normal(0,1)}$$

As t increases, x_t becomes closer to pure Gaussian noise.

The model then takes the noisy image and predicts the noise instead of directly predicting the original image. It receives both the noisy image and the timestep, which is embedded using linear layers. The architecture uses a combination of **residual blocks** and **attention blocks** to compute attended feature maps. The temporal embedding is introduced via concatenation, followed by residual connections to prevent information loss and ensure stable training. Inputs are normalized using **group normalization layers** instead of batch normalization to prevent gradient overflow and avoid batch dependency. Additionally, **SILU activation functions** are used to preserve negative values rather than squashing them completely.

The architecture is organized into **down, middle, and up blocks**. Down blocks perform downsampling, middle blocks act as a transition phase, and up blocks perform upsampling. The model learns the image structure from the noisy input step by step.

This incremental approach requires longer training time but produces higher-quality images with less blur because the steady steps and temporal embeddings guide the model's decisions beyond the sampled noise. Due to the complexity of the architecture, more trainable parameters are required, which increases training time and model size. However, with sufficient high-quality data, training time becomes less of an issue, and the generated images exhibit significantly reduced blur.

- **Reverse process (denoising to reconstruct the image):**

  The model predicts the noise in $x_t$and removes it to approximate x_{t-1}:

$$x_{t-1} \approx x_t - \text{predicted noise}$$

- **Training objective:**

  The model is trained to **minimize the difference** between the real noise added and the noise it predicts:

$$\text{Loss} = || \text{ actual noise} - \text{predicted noise } ||^2$$

Dataset choices:

- **MNIST Dataset**

Dataset Details:

- **Size:** 70,000 images of handwritten digits.
- **Description:** 28×28 pixels.
- **Channels:** Single grayscale channel.

First, the MNIST dataset was used as a baseline to evaluate the efficiency and performance of the models and to observe visual differences on a simple dataset with small image size (28×28, reshaped to 32×32) and a single grayscale channel. The VAE was trained with different beta values to visualize how beta affects the blur in generated images and to monitor the overall training time and convergence for both VAE and DDPM architectures.

- **CIFAR10 Dataset**

Dataset Details:

- **Size:** 60,000 images of 10 classes.

- **Description:** 32×32 pixels.
- **Channels:** 3 colour channels (RGB).

Next, the CIFAR10 dataset was used to go beyond the constrained digits of MNIST and to test the reproducibility and creativity of the models on more complex images. CIFAR10 has diverse structures and multiple class labels, which allows for a better assessment of the models' performance in real-world scenarios. The main challenge was the low resolution (32×32) and complex structures, making quantitative comparisons harder.

- **Oxford Flowers Dataset**

Dataset Details:

- **Size:** 8,189 images of 102 flower categories.
- **Description:** Originally high-resolution (~224×224 pixels), resized to 32×32 for training.
- **Channels:** 3 colour channels (RGB).

Finally, the Oxford Flowers dataset was used to address limitations from CIFAR10. The dataset contains high-resolution images (originally ~224×224pixels), which were resized to 32×32 to avoid memory overflow during DDPM training. Despite the reduced resolution, the dataset is less complex than CIFAR10 in terms of the number of classes, making it suitable for visually evaluating the quality of generated images while still being challenging enough for model generalization.

Implementation steps:

- **Classifier Model Training (for Inception Score):**

  1. A classification model (ResNet-18) was trained separately on each of the three datasets (MNIST, CIFAR10, Oxford Flowers).
  2. The model predicts class probabilities $p(y \mid x)$ and is later used to compute the Inception Score of generated images.
  3. This ensures the Inception Score reflects how confidently and diversely the generative model produces samples resembling the dataset classes.

- **Variational Autoencoders (VAEs)**

  1. **Define VAE Loss Function:**

     - Implemented a flexible loss function that takes a **beta parameter** for controlling the trade-off between reconstruction quality and latent space regularization.
     - The loss function can choose between **Mean Squared Error (MSE)**, **L1 loss**, or **Binary Cross Entropy (BCE)** depending on the data type and distribution assumptions.
     - KL divergence is included to ensure the latent space approximates a Gaussian distribution.

  2. **VAE Model Architecture:**

     - The **VAE class** includes an **encoder**, **decoder**, and **mu/logvar layers**.
     - **Encoding:** The input image is passed through the encoder, flattened, and then mapped to **mu** and **logvar** layers.
     - **Reparameterization:** The latent vector $z$is computed from **mu** and **logvar** using the reparameterization trick to allow backpropagation.
     - **Decoding:** The latent vector z is reshaped into a tensor suitable for the decoder, which reconstructs the image to match the original input dimensions.

  3. **Training Loop:**

     - Compute the VAE loss (reconstruction + KL divergence) for each batch.
     - Perform **backpropagation** to update model weights using the chosen optimizer.

  4. **Training Setup:**

     - Load the chosen dataset (e.g., MNIST, CIFAR10, Oxford Flowers) and specify beta values for the loss.
     - Define training parameters: **learning rate**, **number of epochs**, **optimizer (Adam)**, **learning rate scheduler**, and whether to use **early stopping**.
     - Optionally, resume training from a previously saved **checkpoint** to continue training or fine-tune.

  5. **Evaluation:**

- Assess the trained model using **Frechet Inception Distance (FID)** and **Inception Score** to quantitatively evaluate image quality and diversity.

6. **Sample Generation:**

   - Implement a function to generate new images from the trained VAE by sampling from the latent Gaussian distribution.
   - Default is **100 generated samples**.

- **Denoising Diffusion Probabilistic Models (DDPMs)**

1. **Define the Noise Scheduler Class:**

   - The scheduler handles both the **forward and backward diffusion processes**.
   - During the **forward pass**, it adds noise to the image using a certain timestep.
   - During the **backward pass**, it denoises the image using the computed mean and variance for the current timestep to avoid sudden changes to the original image.

2. **Define the Temporal Embeddings Function:**

   - Takes a **timestep** t and the **latent dimension**.
   - Forms a vector where half of the dimensions are filled with **sine** and the other half with **cosine** values to get the sinusoidal embeddings.
   - The shape of the vector matches the **batch size** of the input and projects it to obtain **sinusoidal embeddings**, which are later used to condition the network on the current timestep.

3. **Define the Downsampling Block Class:**

   - Composed of a **stack of residual blocks** that can be repeated according to a user-specified number of layers. More layers increase computation but can improve image quality.
   - Each downsample block contains:
     - **Residual Block:** group normalization → SiLU activation → convolution layer. The output is added to the temporal embeddings

and passed through the next residual block. The final output is **added to the input** (skip connection).

- **Attention Block:** group normalization → multi-headed self-attention. The output may optionally be downsampled.

- Number of channels generally **doubles** and spatial resolution **halves** to balance computational efficiency with feature preservation.

4. **Define the Middle Block Class:**

- Similar logic to the downsample block.
- Applies a **residual block**, followed by a **sequence of attention and residual blocks** (number of layers user-specified).
- Output is upsampled to be concatenated with the corresponding downsampled features for the upsampling stage.

5. **Define the Upsampling Block Class:**

- Performs the opposite of downsampling:
  - **Upsample** if specified.
  - Apply **residual blocks** and **attention blocks**.
- Each layer receives the output from the previous layer **concatenated** with the corresponding feature maps from the downsampling path (skip connections).

6. **Define the Full U-Net Architecture:**

- Projects the sinusoidal temporal embeddings using linear layers.
- Combines all blocks into a **modular U-Net**:
  - Input image is noised.
  - Noised image passes through the downsampling blocks → middle block → upsampling blocks.

        - Feature maps from downsampling are merged via skip connections.
        - Final output predicts the **noise residual**.

7. **Training Function:**

    - Inputs: **dataloader, optimizer, number of epochs, early stopping, learning rate scheduler, noise sampling parameters**.
    - For each batch:
        - Sample random noise and time step vectors.
        - Noise is added to the image using the **forward scheduler**.
        - The noisy image and timestep are passed through the U-Net model to get the time embeddings and predict noise.
        - The model predicts the noise for the reverse process.
        - **Mean Squared Error (MSE)** is computed between predicted noise and true noise from the forward pass.
        - Optimizer updates the network weights.
        - Checkpoints are saved, and a learning rate scheduler is applied to maintain stable training.

8. **Evaluation Function:**

    - Computes metrics such as **FID** and **Inception Score** to assess image quality and diversity.

9. **Generation Function:**

    - Generates new samples by starting from pure noise and iteratively denoising using the trained DDPM model.

- The process uses the learned reverse diffusion steps and temporal embeddings to produce final images.

Explanations for Architecture and Training Choices:

1. **Optimizer: Adam**
   - **Choice:** Adam optimizer was chosen for both VAE and DDPM training.
   - **Reasoning:** Adam combines momentum and adaptive learning rates, which stabilizes training on deep networks. In our experiments, weight decay (AdamW) was not strictly necessary because overfitting was controlled via early stopping and dataset size.

2. **VAE Reconstruction Loss Function: MSE, L1, or Binary Cross-Entropy (BCE)**
   - **Choice:** Flexibility to select MSE, L1, or BCE based on the dataset and output type.
   - **Reasoning:**
     - **MSE:** Best for continuous image outputs; pushes outputs toward the mean → c an produce blurrier images in VAEs.
     - **L1:** Less sensitive to outliers → can produce sharper images than MSE.
     - **BCE:** Used when outputs are binary or normalized [0,1]; common in grayscale datasets like MNIST.
   - **Benefit:** Allows experimenting with reconstruction quality vs latent space regularization.

3. **UNet Architecture for DDPM (Inspired by Stable Diffusion / Hugging Face)**
   - **Choice:** Modular UNet with downsample, middle, upsample blocks, residual and attention layers.

- **Reasoning:**
  - UNet allows **multi-scale feature extraction** via skip connections.
  - Attention layers capture **long-range dependencies**, improving global image consistency.
  - Residual connections prevent **vanishing gradients**, stabilizing the reverse diffusion process.
  - Temporal embeddings allow the model to know the **noise level at each timestep**.
- **Justification:** UNet is the backbone of modern diffusion models like **Stable Diffusion**, proven effective in generating high-quality, diverse images.

4. **Early Stopping**
   - **Choice:** Early stopping is applied based on validation loss or training metrics.
   - **Reasoning:**
     - Prevents overfitting, especially important when using high-capacity models like DDPM.
     - Saves computational resources by stopping training once improvement plateaus.
     - Ensures that the best model checkpoint corresponds to minimal loss, improving generalization.

5. **Scheduler**
   - **Choice:** Learning rate scheduler (e.g., ReduceLROnPlateau) used.
   - **Reasoning:**
     - Adjusts learning rate dynamically based on training progress.
     - Helps the optimizer converge more smoothly and avoids overshooting minima.
     - Particularly useful for DDPMs because the reverse process is sensitive to small errors in noise prediction.

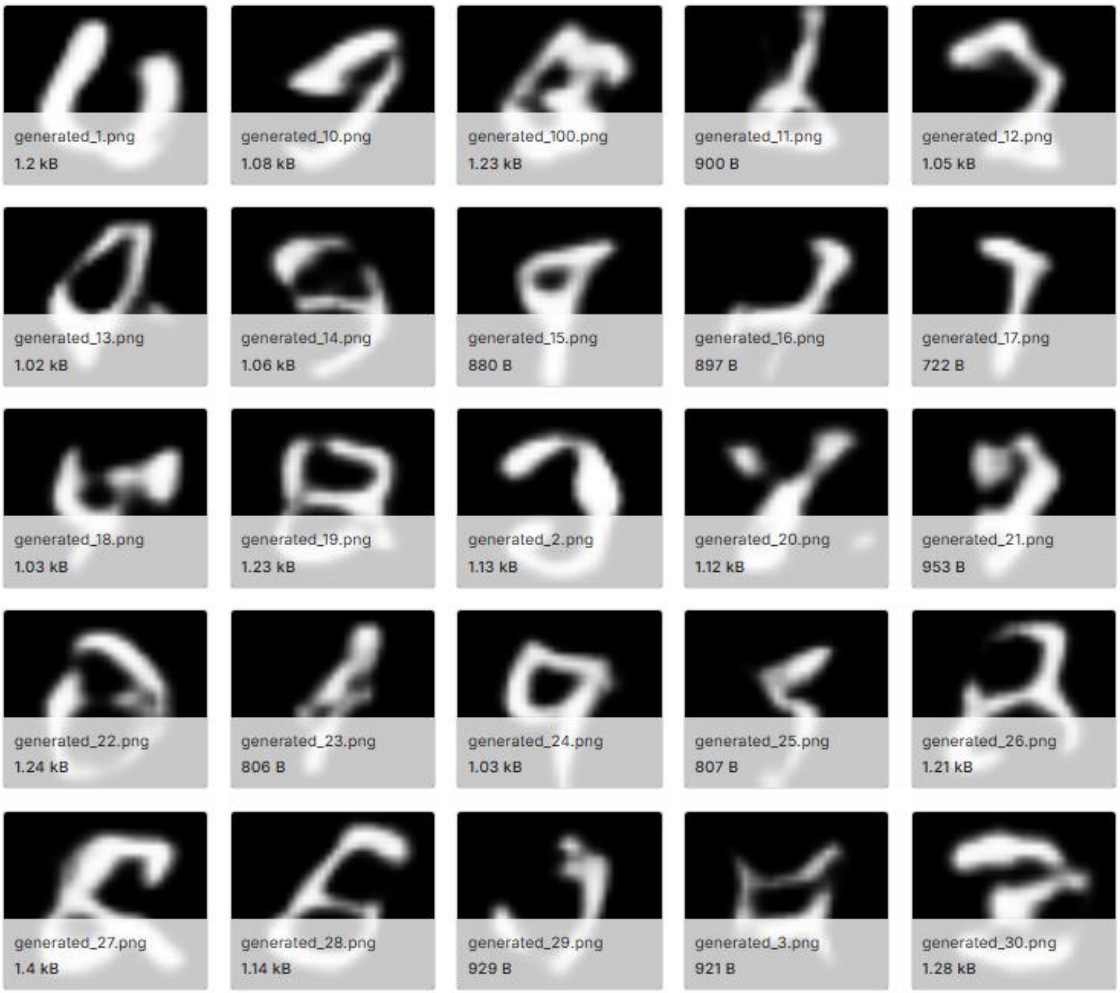
6. **Metrics: FID and Inception Scores**

- **Choice:** Used to quantitatively evaluate generative model performance.
- **FID (Fréchet Inception Distance):**
  - Measures similarity between the distribution of generated images and real images.
  - Lower FID → generated images closely match real image distribution (more realistic and diverse).
- **Inception Score (IS):**
  - Evaluates image quality and diversity based on a pretrained classifier.
  - Higher IS → images are both high-quality and diverse across classes.
- **Justification:** These metrics allow **objective comparison** of VAE vs DDPM, complementing qualitative visual assessment.

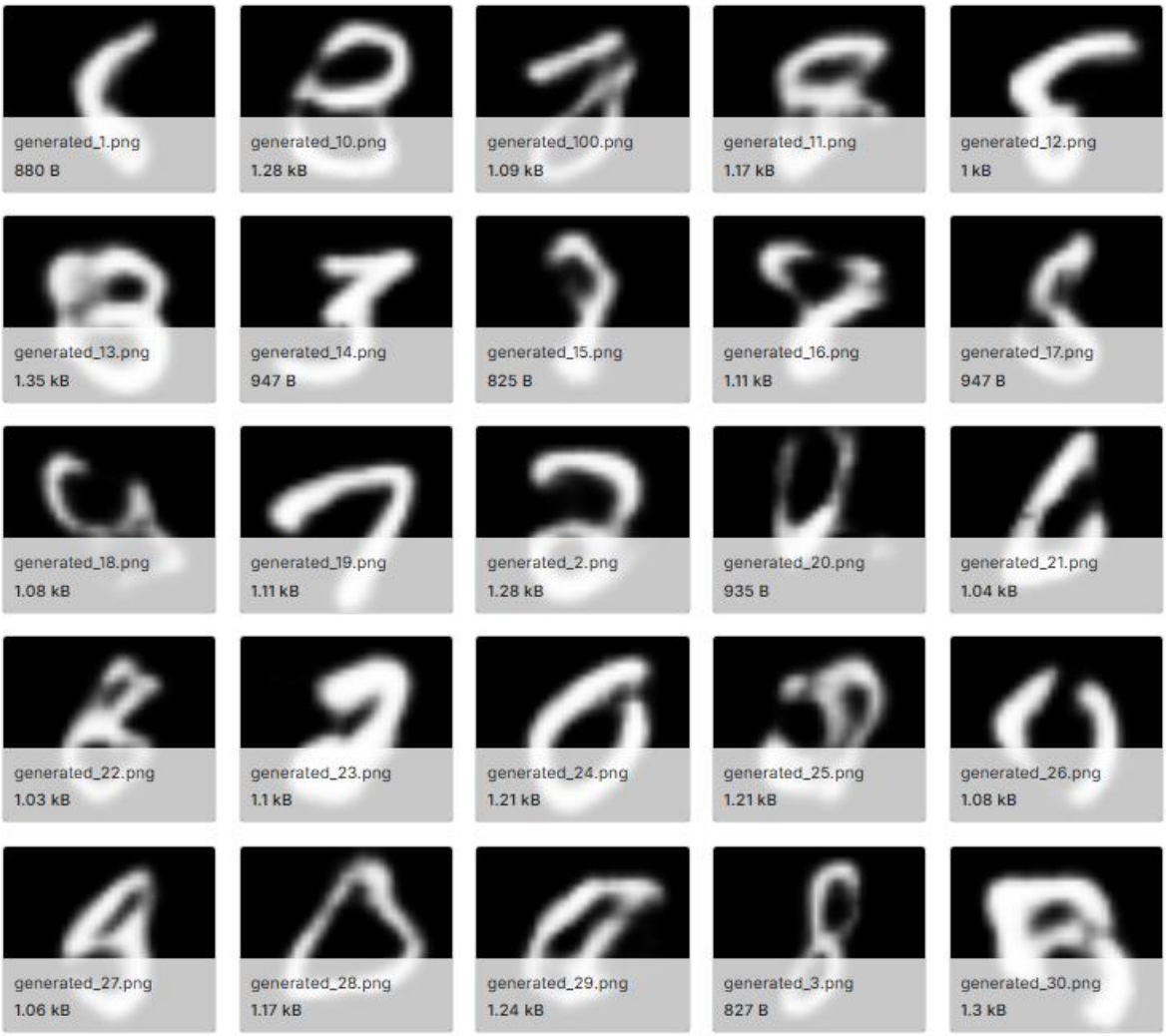
Experimental Results & Explanations:

- **MNIST Dataset**

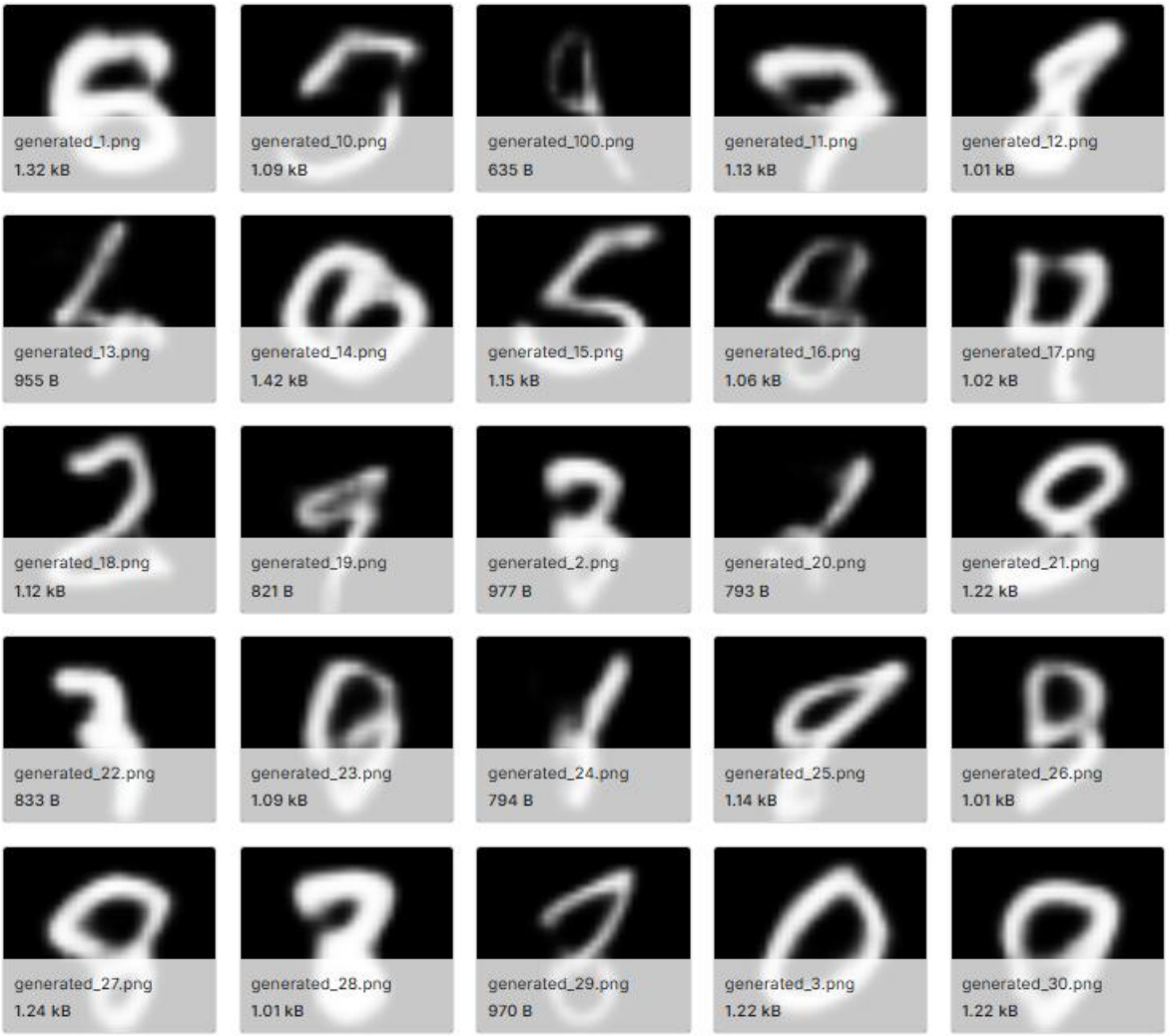
Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	164	108.7340	360.502	4.738 +/- 0.7419	**0.1**



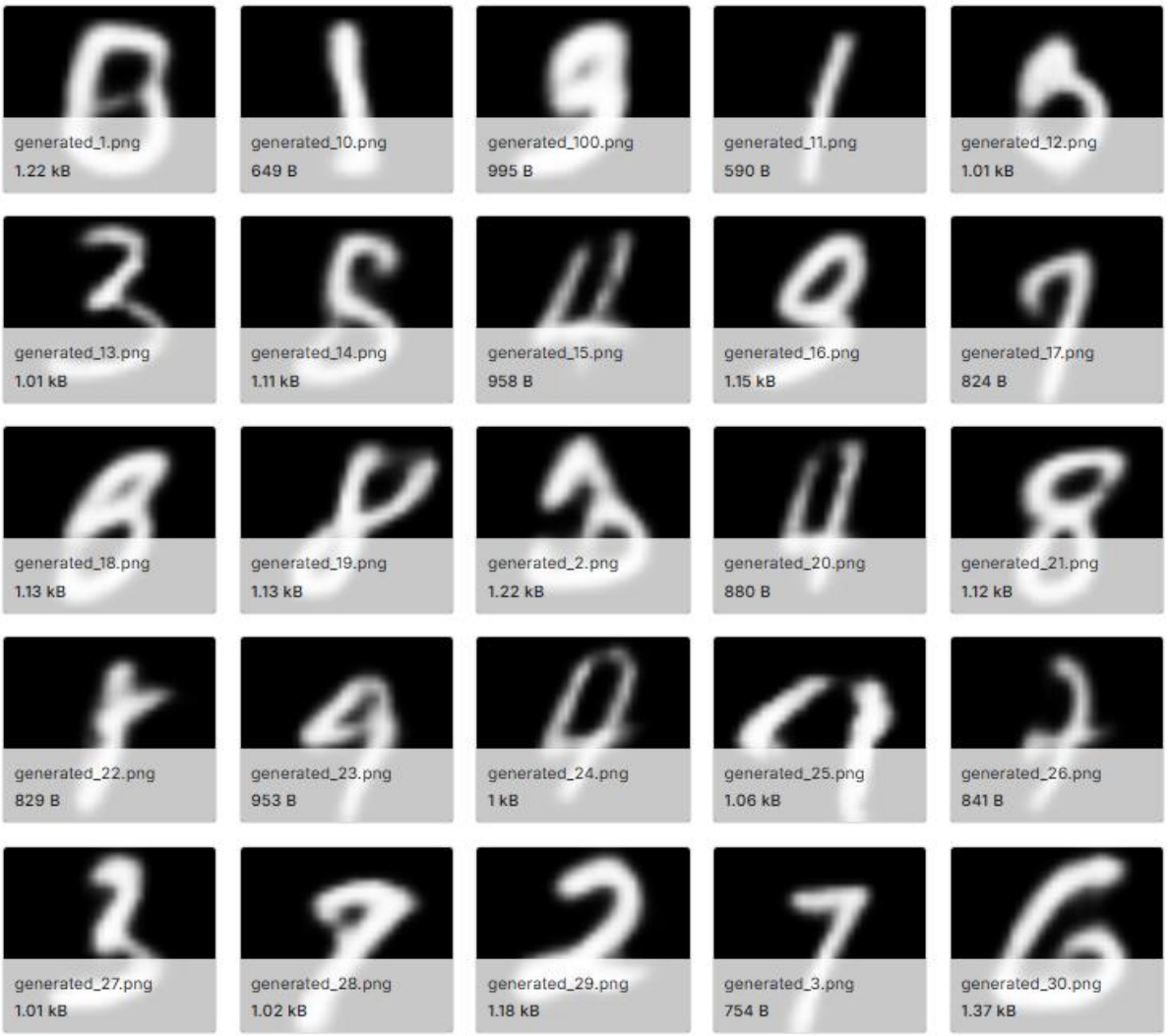
Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	185	118.9870	351.394	4.9076 +/- 0.843	**0.3**



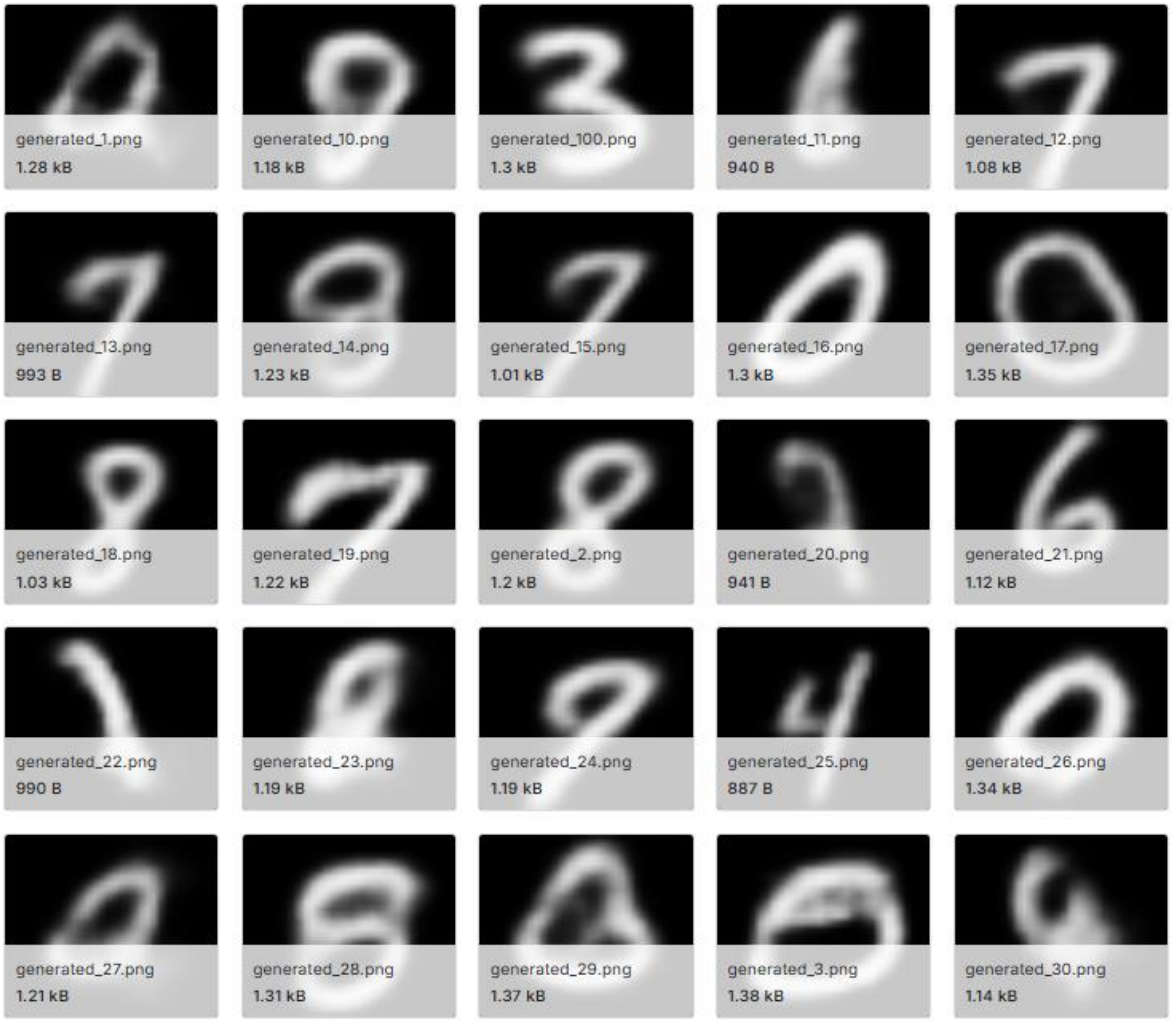
Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	214	126.1850	346.837	5.2535 +/- 0.968	**0.5**



Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	219	139.1379	342.085	5.6207+/- 0.9189	**1**



Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	113	195.2625	352.915	5.2910 +/- 0.536	**5**



Comments:

From investigating the generated images, it is clear that as the **beta value increases**, the reconstructed images better match the distribution of the training dataset. For example, when **β = 5**, the shapes of the reconstructed digits closely resemble the original MNIST digits. Additionally, as beta increases from **0.1 to 1**, the **FID score decreases**, which aligns with expectations since FID measures the difference between the distributions of real and generated images, moreover the **Inception score** decreases which further assures the claim regarding the reality of constructed images.

However, the VAE model appears to **repeat certain structures**, showing less diversity—it tends to generate more circular digits such as **0, 6, and 8**. The effect of beta on blurriness is **not very noticeable** for MNIST, likely due to the simplicity of the dataset. Overall, images generated by the VAE remain **noticeably blurry**, regardless of the configuration, which is consistent with the known limitations of VAEs when using MSE-based reconstruction loss.

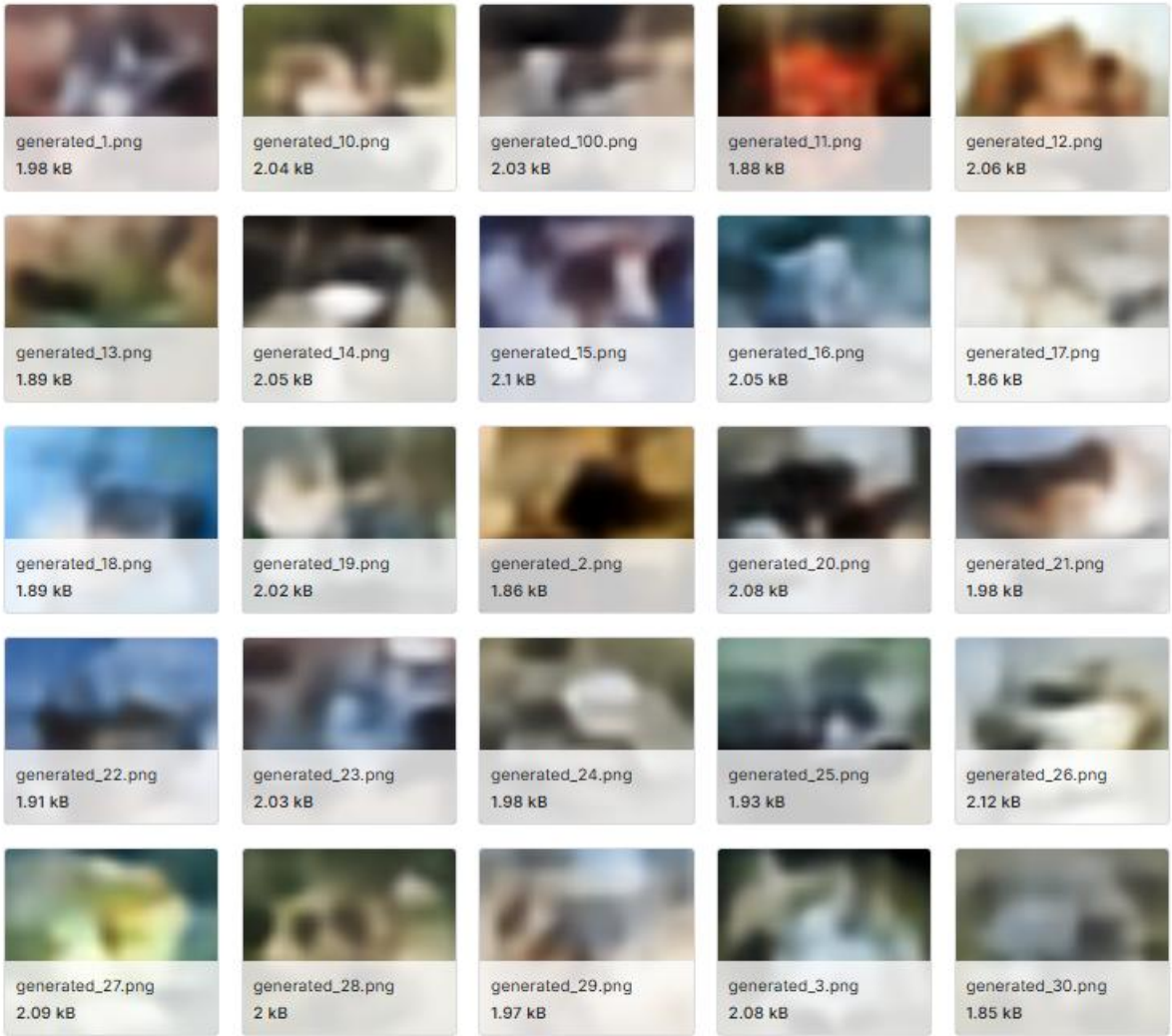
Model	Epochs	Train Loss	FID	Inception Score	Beta
DDPM	10	0.0003	74.295	5.1098 +/- 0.794	_

Comments:

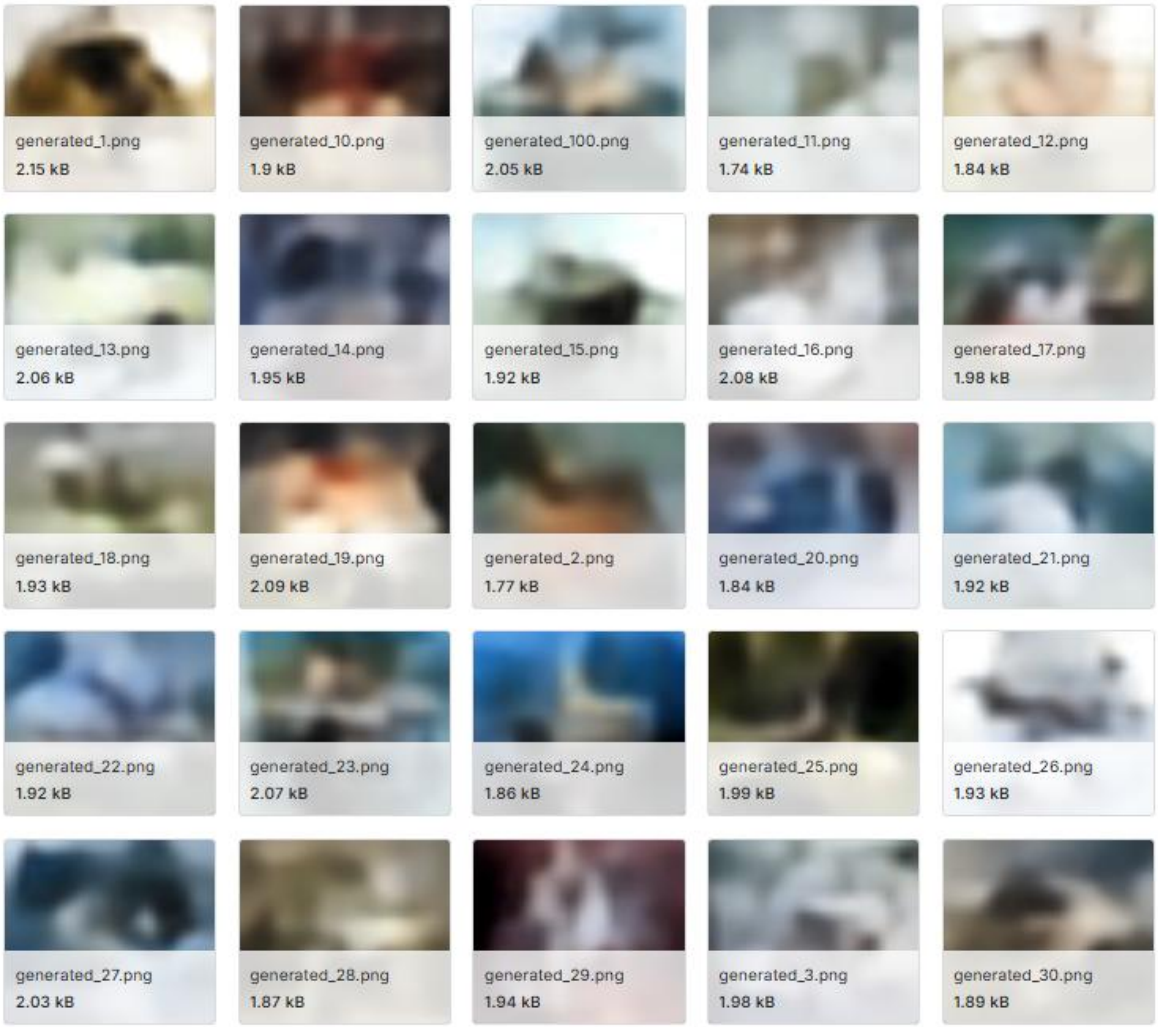
The DDPM model produces **high-quality and sharp images** for MNIST, as expected for such a simple dataset. The **FID score of 74.295** indicates that the generated distribution is closer to the real dataset compared to most VAE configurations. The images are **diverse**, covering all digit classes without repeating particular shapes excessively. Unlike VAEs, the DDPM is able to produce **clear boundaries and well-defined digits** even with relatively few training epochs. Overall, for MNIST, the DDPM outperforms VAE in both **distribution similarity** (FID) and **visual sharpness**, demonstrating its advantage in modelling simple datasets.

- **CIFAR10 Dataset**

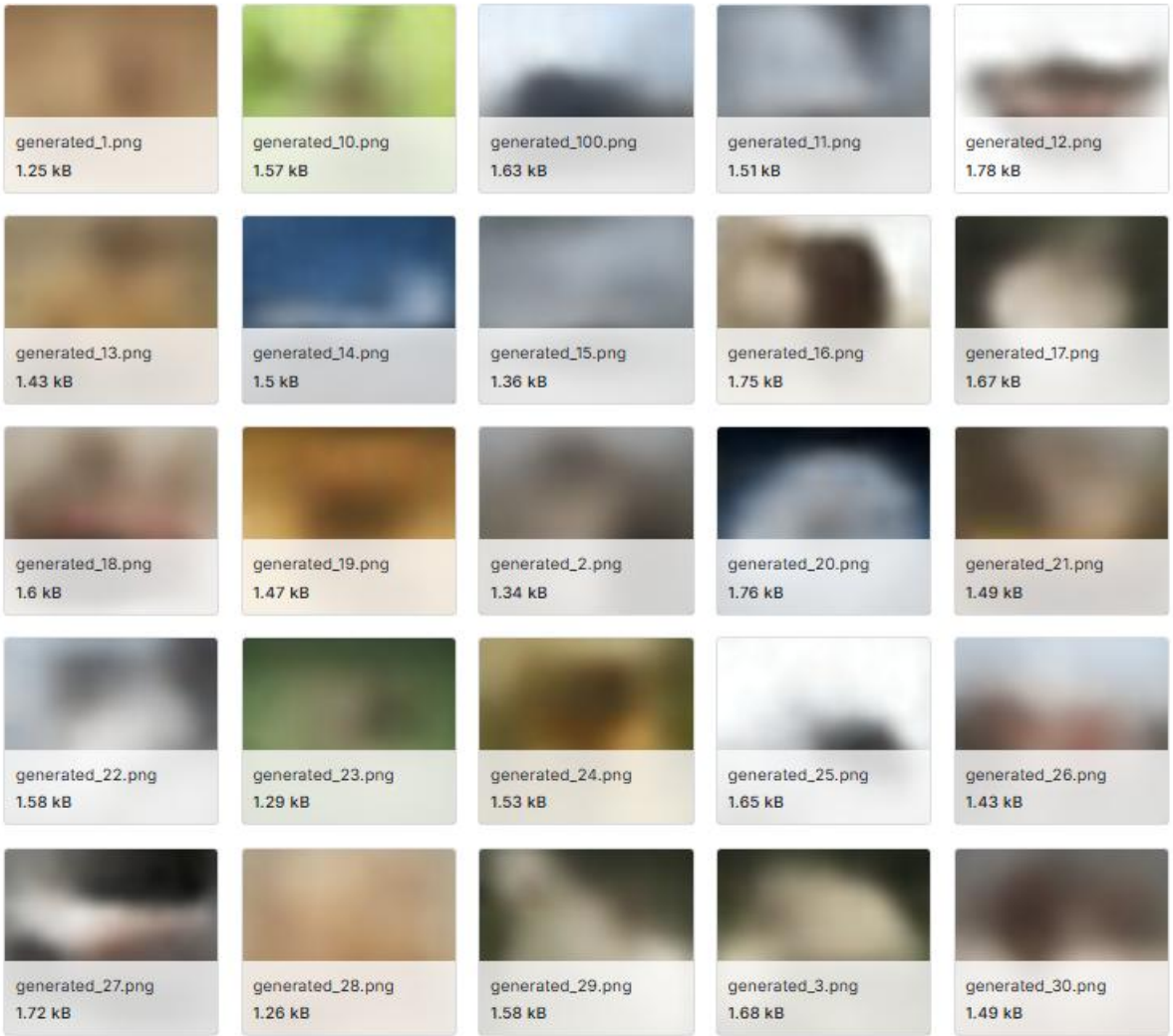
Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	194	109.9439	364.580	2.250 +/- 0.3101	**0.3**



Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	164	134.1314	354.650	2.379 +/- 0.256	**0.5**



Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	117	310.7477	362.678	1.350 +/- 0.095	**5**



Comments:

By investigating the generated images, it is clear that as β increases from 0.1 to 1, the model focuses more on reconstruction, which helps preserve edges and some fine details. In this range, the latent distribution is less strictly constrained to follow a Gaussian prior. As a result, the generated images tend to exhibit relatively "bold" or sharper edges, as the model prioritizes reconstructing visible features. However, due to the complexity of the CIFAR-10 dataset, multiple object features or background elements may still appear, often in a distorted or unclear manner.

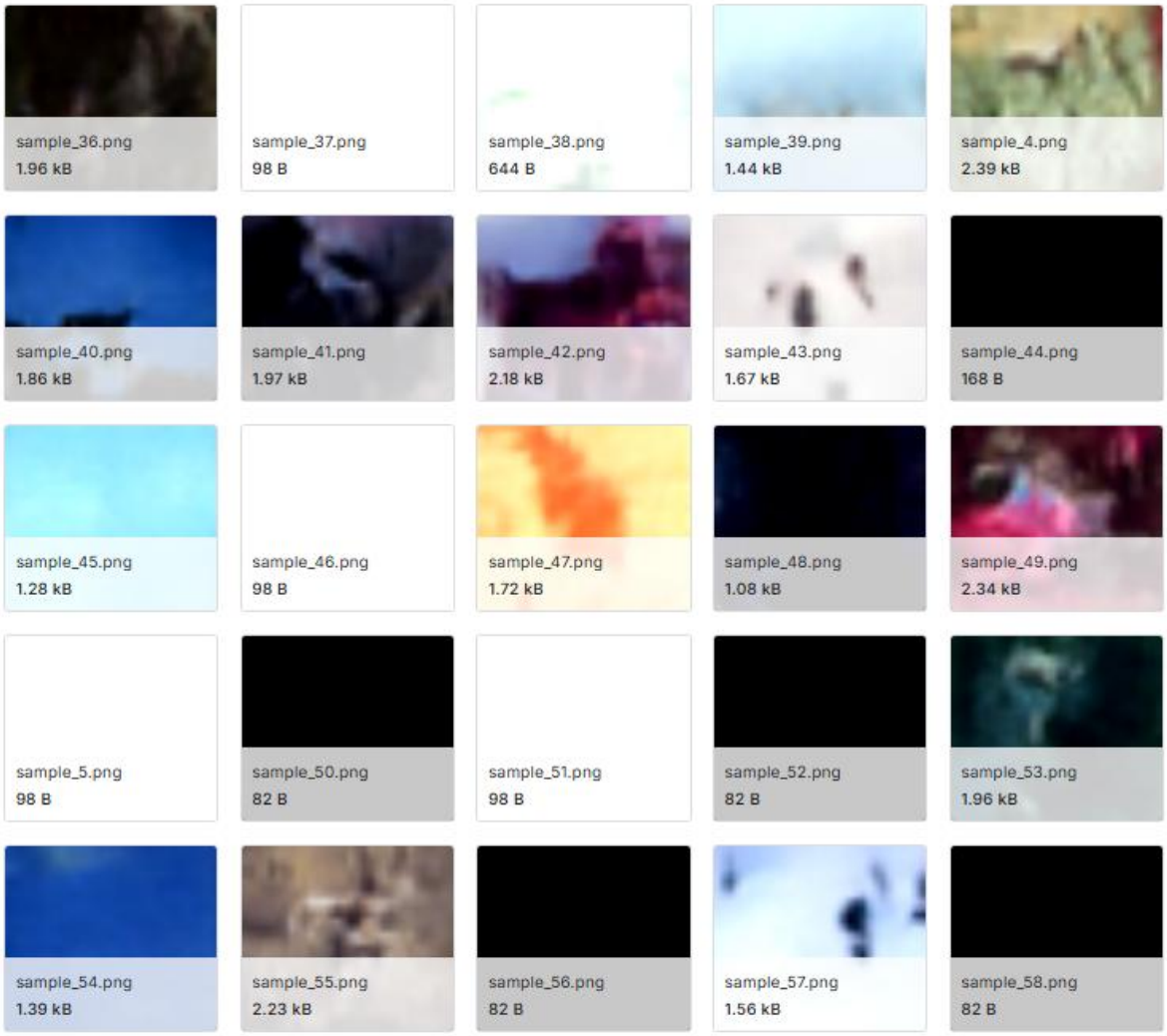
For larger β values (e.g., β = 5), the model prioritizes latent space regularization, where the KL divergence term dominates. This enforces a more structured and Gaussian latent space but comes at the expense of reconstruction quality. Consequently, the model produces smoother and more simplified images, focusing primarily on the main object

category (such as cars, airplanes, or animals), while losing fine-grained textures, background details, and intra-class variations.

Additionally, CIFAR-10 is significantly more complex than datasets like MNIST, as it contains color images with diverse object classes, viewpoints, and backgrounds. This increased complexity makes it harder for the VAE to capture detailed structures, and the effect of blurriness becomes more pronounced.

Overall, the generated images remain blurry across all configurations, which is a known limitation of VAEs using MSE reconstruction loss.

Model	Epochs	Train Loss	FID	Inception Score	Beta
DDPM	15	0.0021	449.544	1.6021 +/- 0.302	_



Comments:

The DDPM model struggles significantly with the CIFAR-10 dataset, as reflected by the high FID score of 449.544 and low Inception Score of 1.6021. The generated images lack clear structure or recognizable object boundaries, suggesting the model has difficulty capturing the complex visual patterns present in the dataset.

This can be attributed to several factors. Unlike MNIST, CIFAR-10 contains colour images with high intra-class variability, diverse backgrounds, and complex textures — all of which demand significantly more model capacity and training data to learn effectively. DDPMs are known to require large, high-quality datasets and extended training to fully leverage their denoising capabilities. With only 15 epochs and 60,000

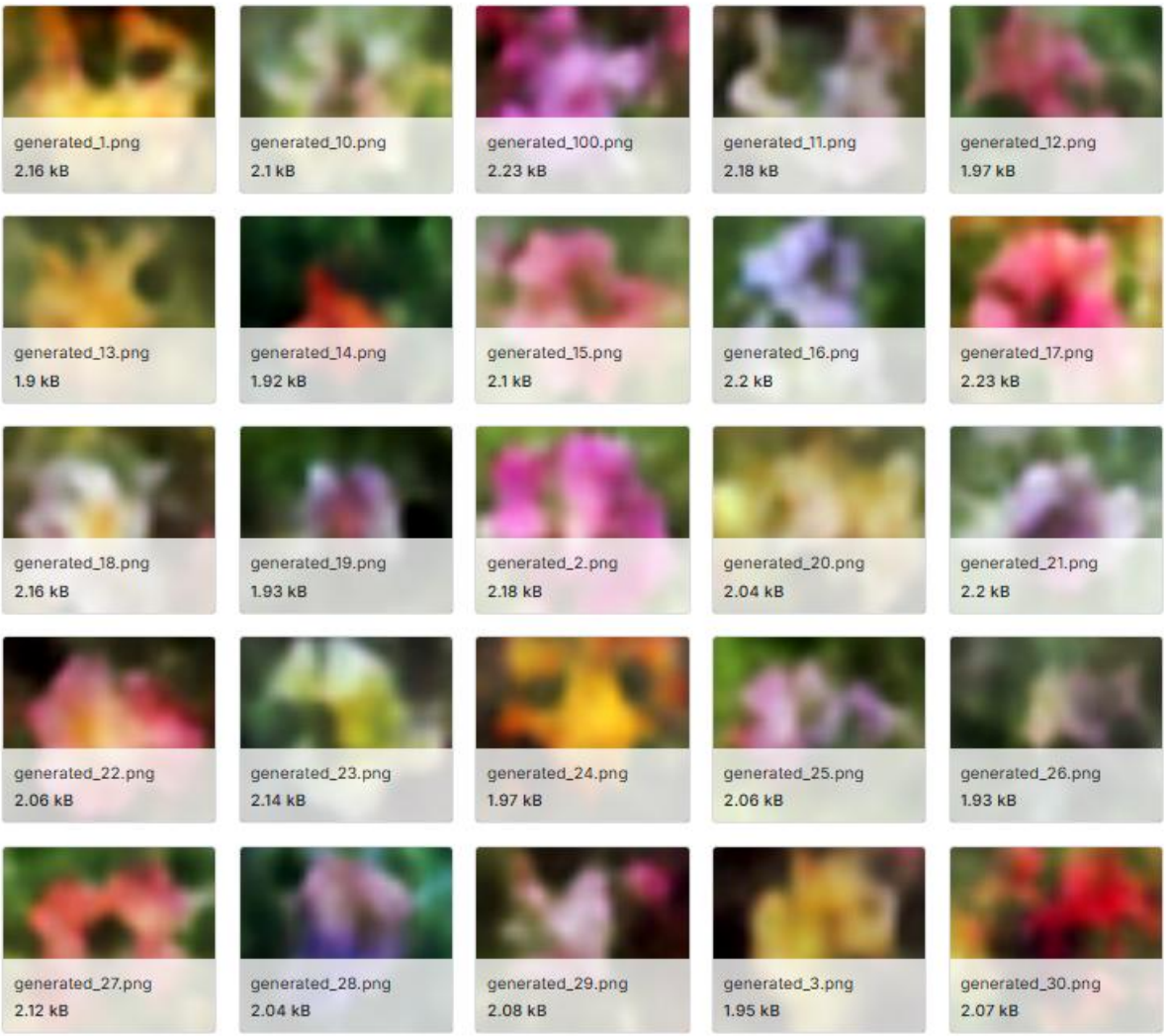
relatively low-resolution images, the model has insufficient exposure to learn the reverse diffusion process reliably.

Furthermore, the 32×32 resolution limits the amount of recoverable detail during the denoising steps, making it harder for the model to distinguish meaningful structure from noise. While the generated samples show some colour patterns vaguely resembling the dataset distribution, they fall short of producing coherent objects.

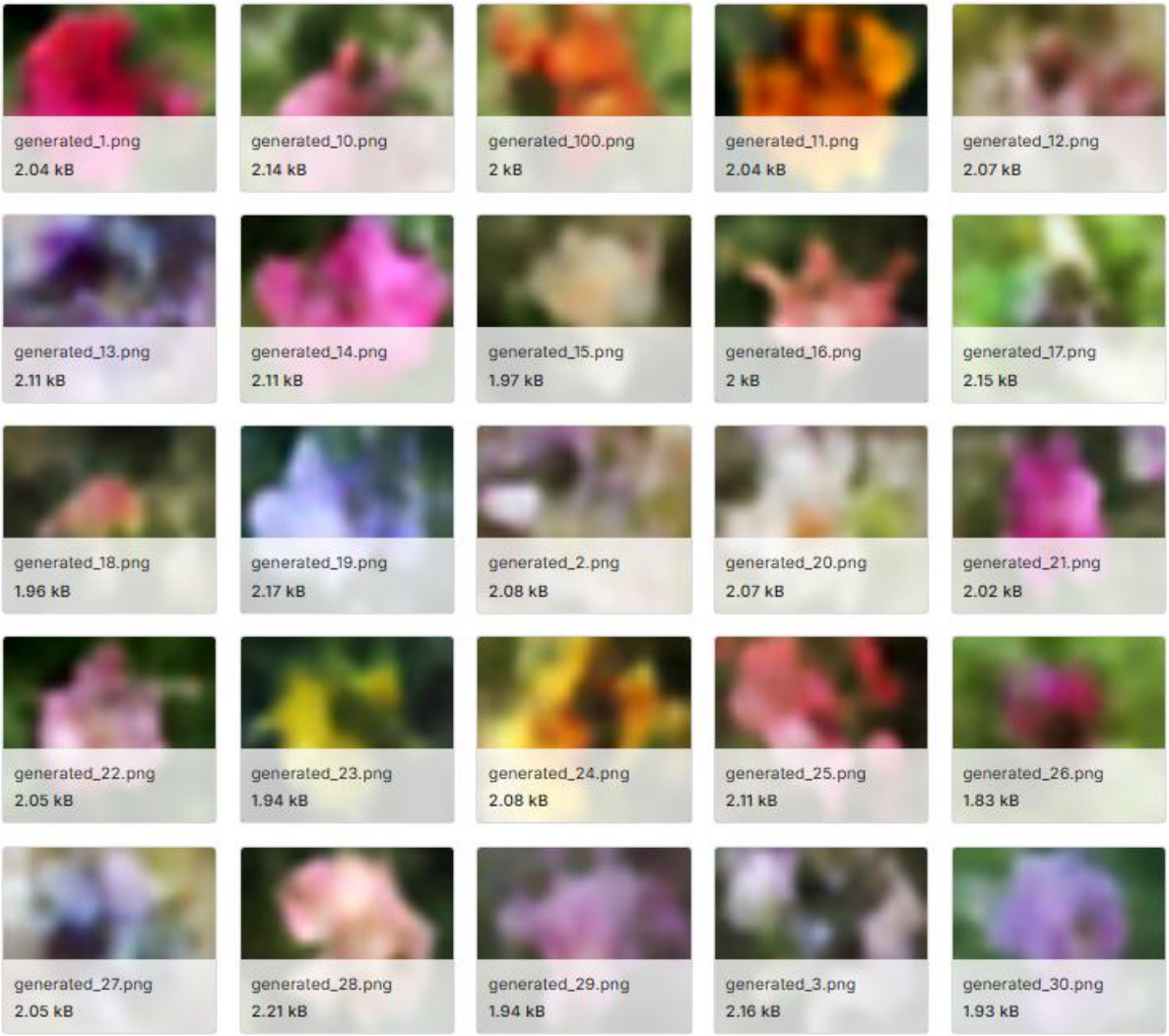
In contrast to the MNIST results — where DDPM clearly outperformed VAE — on CIFAR-10 neither model produces satisfactory outputs, though for different reasons: the VAE suffers from inherent blurriness due to MSE loss, while the DDPM suffers from insufficient training.

- **Oxford Flowers Dataset**

Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	212	124.502	302.729	1.1408 +/- 0.034	**0.3**

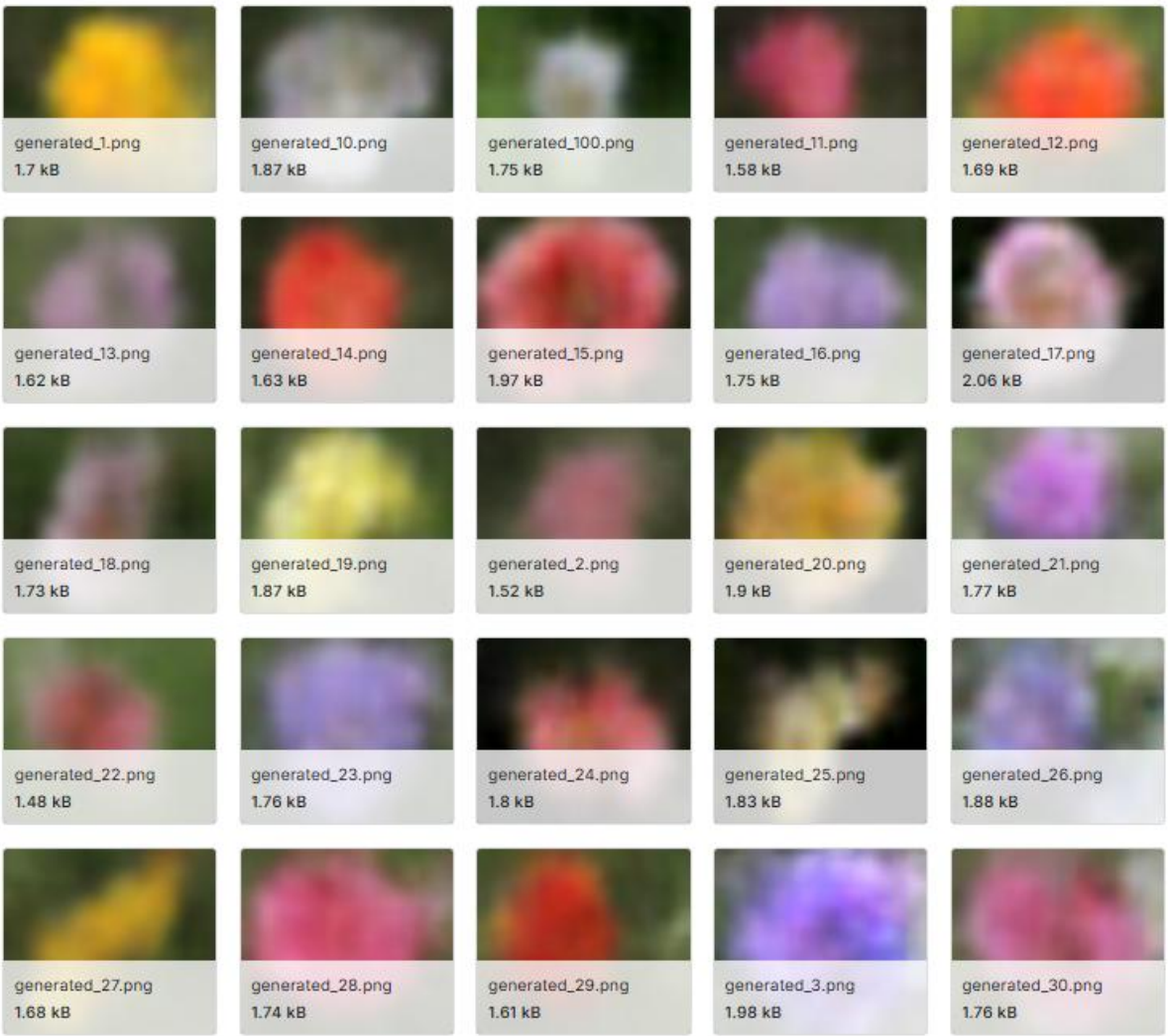


Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	218	150.1791	315.772	1.1678 +/- 0.035	**0.5**



Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	163	197.594	295.922	1.1876 +/- 0.058	**1**

Model	Epochs	Train Loss	FID	Inception Score	Beta
VAE	134	319.738	319.678	1.1719 +/- 0.028	**5**



Comments:

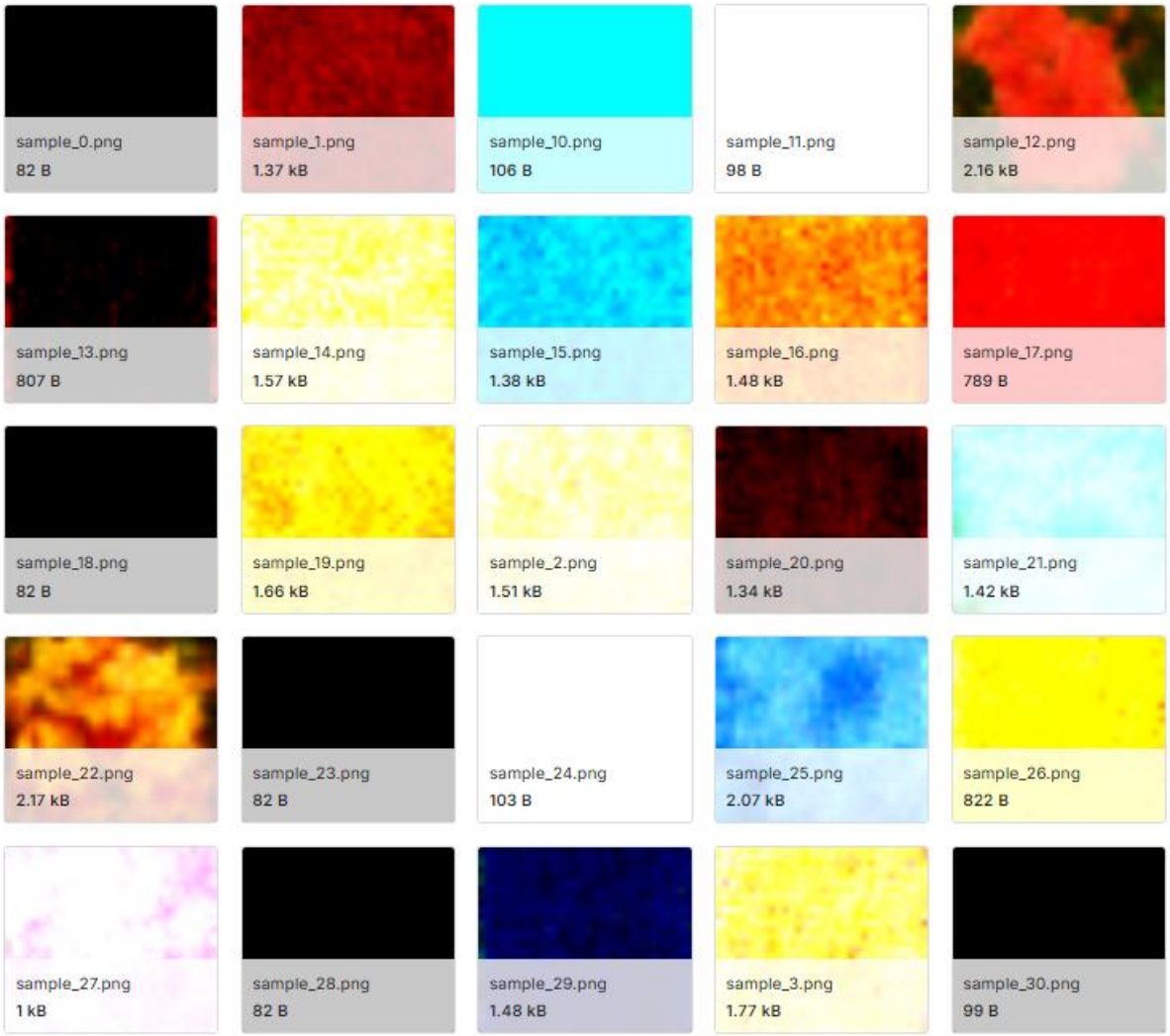
By investigating the generated images, it is clear that as β increases from **0.1 to 1**, the model focuses more on reconstruction, leading to better preservation of edges and fine details. In this range, the latent distribution is less strictly constrained to follow a Gaussian prior. As a result, the images tend to have relatively **"bold" edges**, since the model prioritizes reconstructing exact visual features. Consequently, additional objects or background elements may still appear beyond the main flower, even though the images remain somewhat blurry.

For larger β values (e.g., **β = 5**), the model prioritizes **latent space regularization**. The KL divergence term dominates, forcing the latent space to more closely follow a standard Gaussian distribution. This comes at the cost of reconstruction quality, where the model increasingly **smooths out details** and focuses primarily on the **main object**,

such as the flower. Fine textures, secondary objects, and subtle variations are largely lost, resulting in simpler and more abstract representations.

Additionally, compared to simpler datasets like MNIST, the **Oxford Flowers dataset is significantly more complex** (higher variability in color, texture, and background). This makes the blurriness more noticeable, especially when using MSE-based reconstruction loss in VAEs, which inherently encourages averaging over possible outputs.

Model	Epochs	Train Loss	FID	Inception Score	Beta
DDPM	700	0.0028	377.4357	31.166 +/- 0.030	_



Comments:

The DDPM model produces its worst results on the Oxford Flowers dataset. The generated images show no recognizable flower shapes or structures at all. The images are simply noisy, colourful patches that reflect the bright colours of the dataset but nothing more.

Despite training for 700 epochs, the model fails to generate meaningful images. This is mainly due to the small dataset size, with only 8,189 images spread across 102 flower categories, the model does not have enough examples to learn from. DDPMs are known to require large amounts of data to perform well, and this dataset falls far short of that requirement.

Another contributing factor is the resolution. The original high-resolution images were resized down to 32×32 to avoid memory issues, which removes most of the fine details such as petal shapes and textures. With so little visual information available per image, the model struggles to learn any meaningful structure.

The high **FID score** of **377.4357** confirms this, indicating that the generated images are far from resembling real flowers. The model only seems to have picked up on the general colour distribution of the dataset, which explains why the outputs are colourful but structureless.

Overall, the Oxford Flowers dataset is not well-suited for DDPM training under these conditions. Better results would require a much larger dataset, higher image resolution, or a different approach such as latent diffusion, which is more memory-efficient and better suited for high-resolution image generation.

Conclusions:

Qualitative Analysis: Across all three datasets, the VAE consistently produces blurry but recognizable outputs, while the DDPM produces sharper images when given sufficient data, as clearly seen with MNIST. In terms of diversity, the VAE tends to repeat certain structures — particularly circular shapes in MNIST — whereas the DDPM generates more varied samples. Regarding realism, neither model performs well on complex datasets like CIFAR-10 and Oxford Flowers under the current training constraints, though the VAE at least produces structurally coherent, if blurry, outputs.

Key Takeaways:

- VAEs are simpler, faster to train, and produce decent results even on small datasets, but are fundamentally limited by blurriness due to MSE-based reconstruction loss.
- DDPMs produce sharper and more diverse images, but are highly data-hungry and computationally expensive, making them less practical under constrained settings.
- The beta parameter in VAEs offers a useful trade-off — higher beta values improve distributional alignment at the cost of reconstruction detail.
- Dataset size and resolution play a critical role in DDPM performance, as demonstrated by the stark contrast between the strong MNIST results and the poor Flowers results.

For the Full Implementation Steps: GitHub Repo